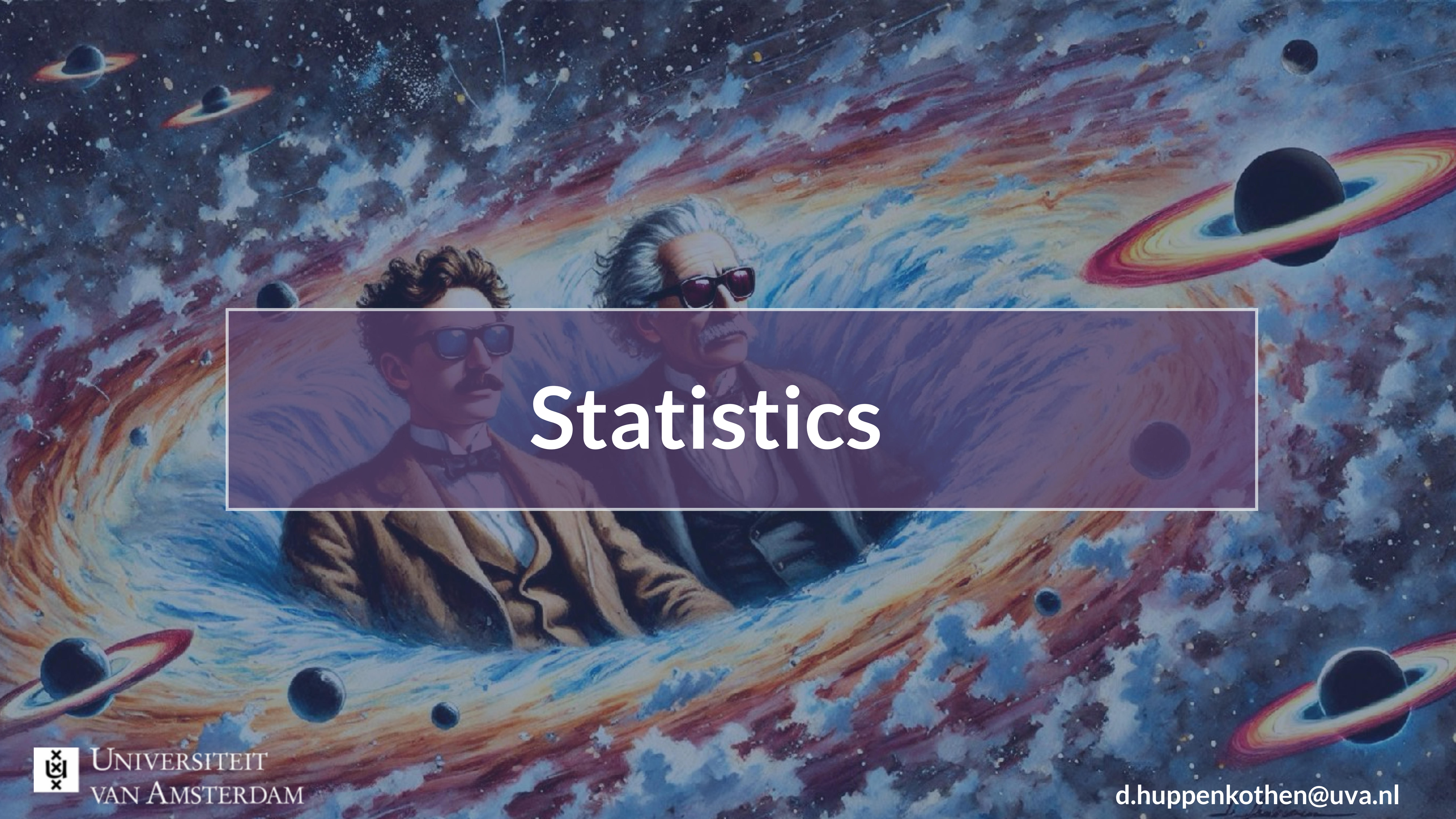
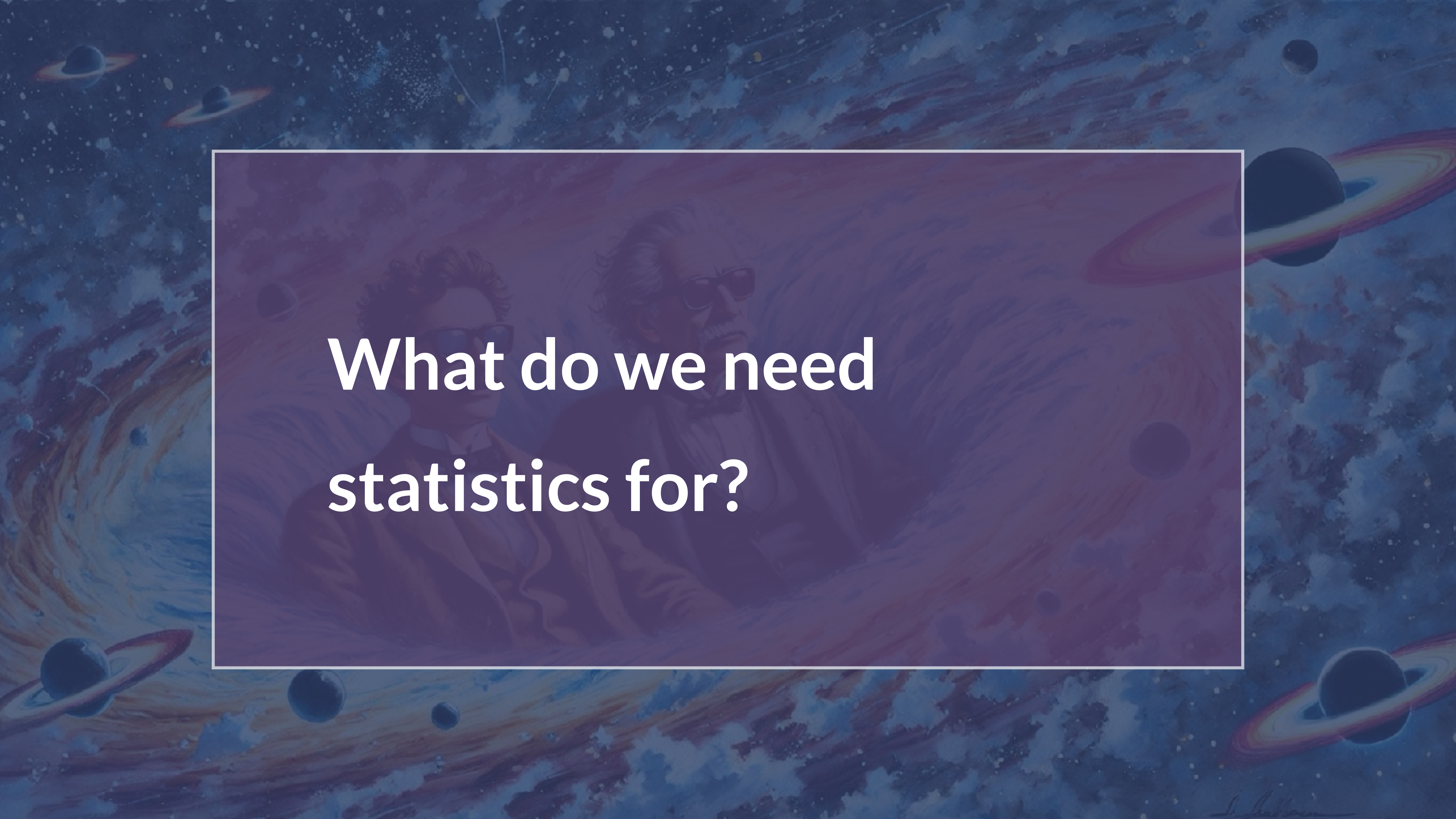


Statistics

Universiteit van Amsterdam

d.huppenkothen@uva.nl



What do we need statistics for?

Where have you encountered statistics in your studies?

[MAKE WOOCLAP FORM]

Where have you encountered statistics in your day-to-day lives?

What is statistics?

What is the recurring theme her?

Data!

- study of uncertainty
- Estimate and summarise properties about data
- Learning from data
- Analysing data
- Collecting data

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Nearly all fields that collect data use statistics: astronomy, biology, chemistry, environmental sciences, forestry, geophysics, law, politics, sports analytics, …

What is statistics?

What is the recurring theme her?

Data!

- study of uncertainty
- Estimate and summarise properties about data
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- Analysing data
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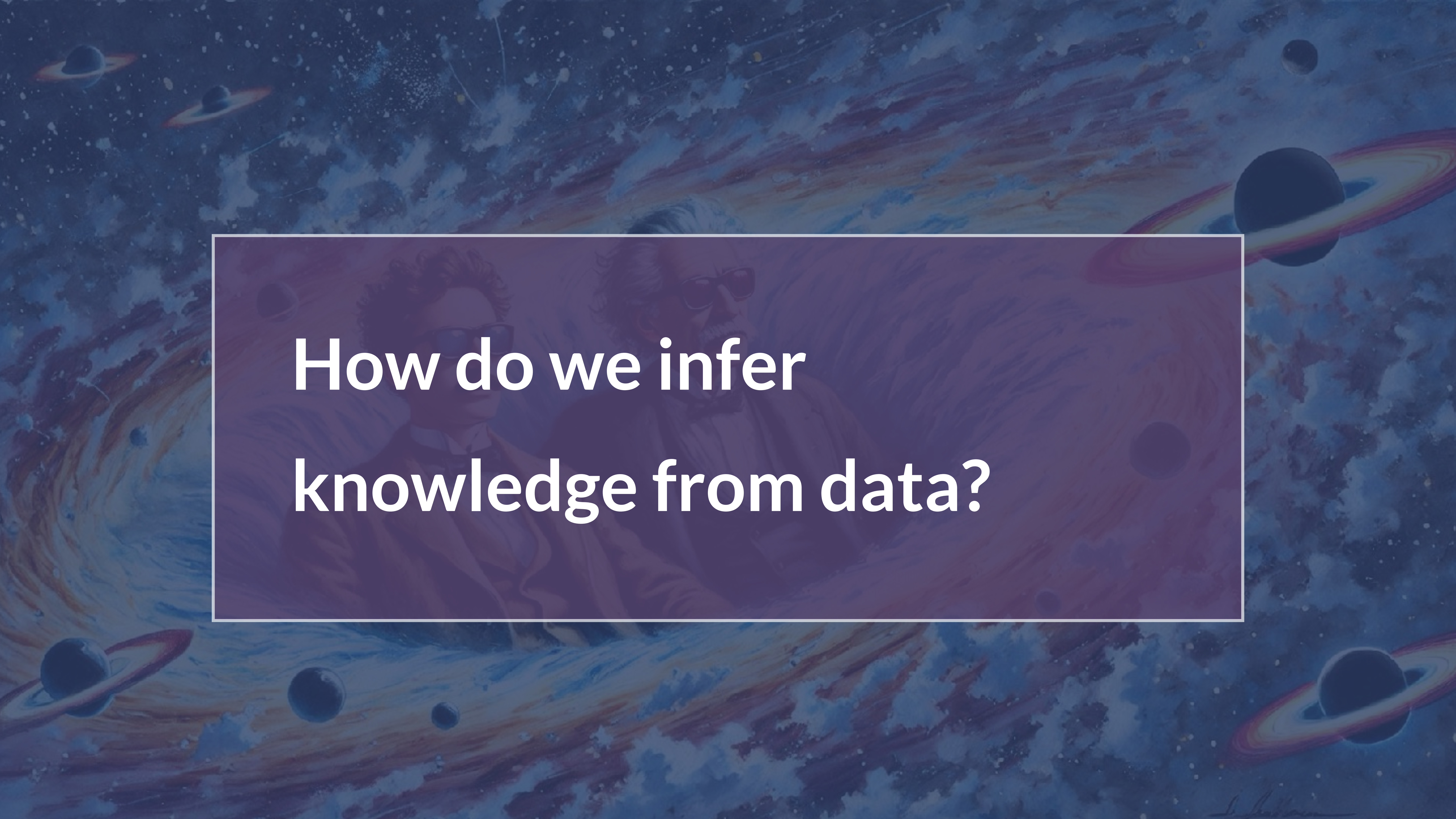
Statistics is used to infer facts, make predictions, and make recommendations, based on data

Nearly all fields that collect data use statistics: astronomy, biology, chemistry, environmental sciences, forestry, geophysics, law, politics, sports analytics, …


How do we infer
knowledge from data?

Course Goals

Explain the foundational concepts of statistics, including probability, randomness, statistical distributions

Formulate a statistically robust data analysis process that considers the limitations of the statistical methods used and the data to be analysed

Choose an appropriate statistical test for a given problem, and employ hypothesis testing to make inferences about data.

Apply Bayesian probability theory to scientific problems.

Develop simple computer programmes to analyse data and apply statistical testing in practice.

Critically interpret the results of a statistical test, including an evaluation of the appropriateness of the test, and its assumptions and limitations, in comparison to the data

Evaluate scientific results in the academic literature within physics/astronomy and in popular media.

Course Goals

This is a course in how to think critically about data and its use (and misuse)!

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My goal: make learning statistics not suck!

Interlude: Logistics

Logistics:

- Central landing site: the course Canvas homepage
- Course materials: website (see Canvas page) and slides (on Canvas): you'll need both!
- In-course information: e-mail (via Canvas)

Structure:

- Lecture/Tutorials: Mondays, Wednesdays, Fridays in weeks 1-3. Mon, Tue in week 4.
- Each ~1/2 interactive lectures, 1/2 tutorial time to work on assignments + practice questions
- Assessment: assessment through 2 statistical programming Assignments (50%) + exam (50%)
- Exam (ADD TIME/LOCATION/DATE) is worth 50% of grade. Short questions, aiming to test theoretical understanding of the topics covered. Example questions will be discussed in the tutorials.
- Important: Exam has a resit but there are no resits for assignments. However, 1d/2d-late submission is possible for assignments with 20%/40% grade penalty
- Check the plagiarism rules and guidance, also rules for AI use!

But can't we just do this with AI?

Fun fact: LLMs are probabilistic models, so understanding statistics helps you make sense of how LLMs work! 🤓

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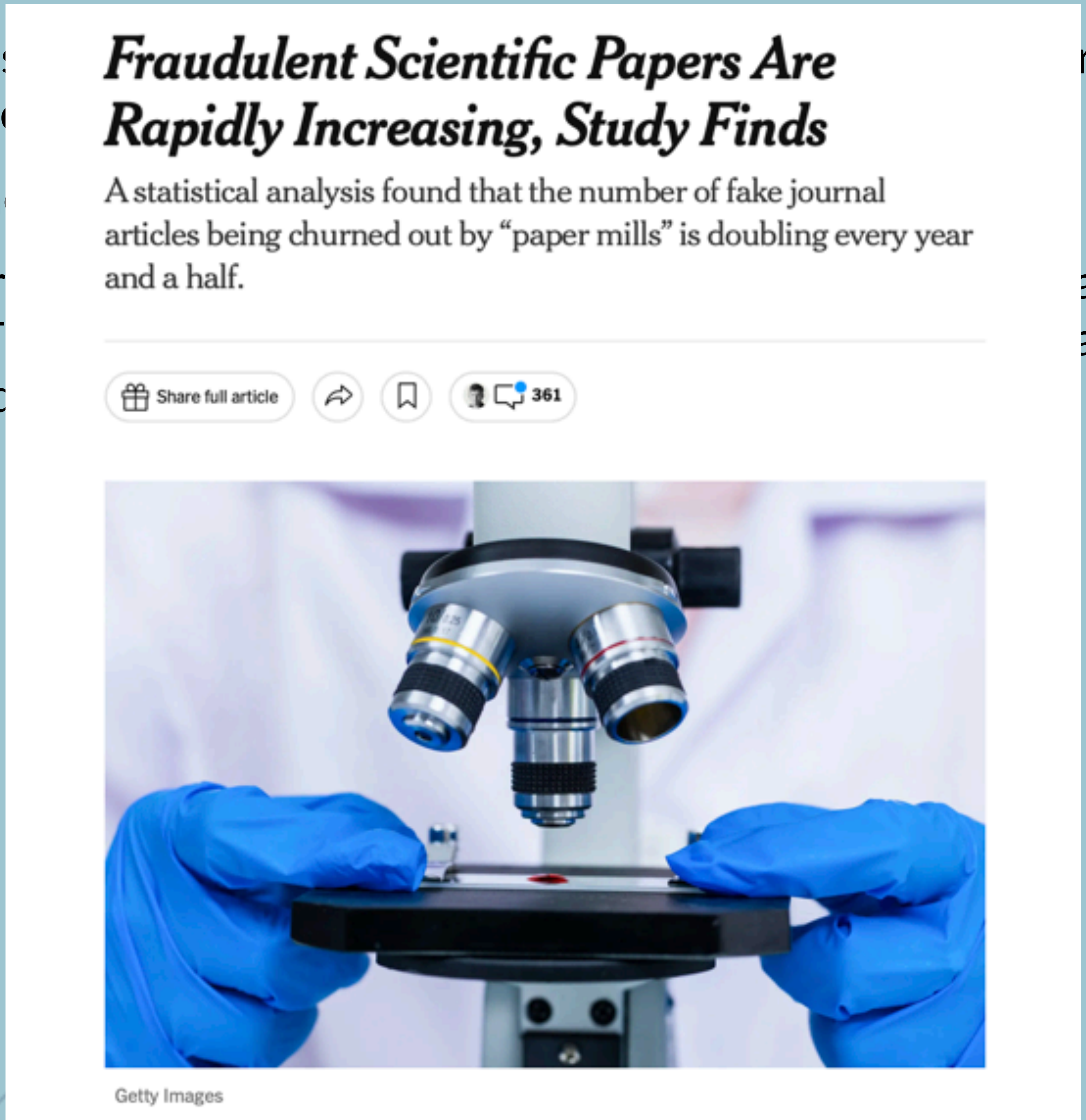

Fraudulent Scientific Papers Are Rapidly Increasing, Study Finds

A statistical analysis found that the number of fake journal articles being churned out by "paper mills" is doubling every year and a half.

Share full article

361

Getty Images

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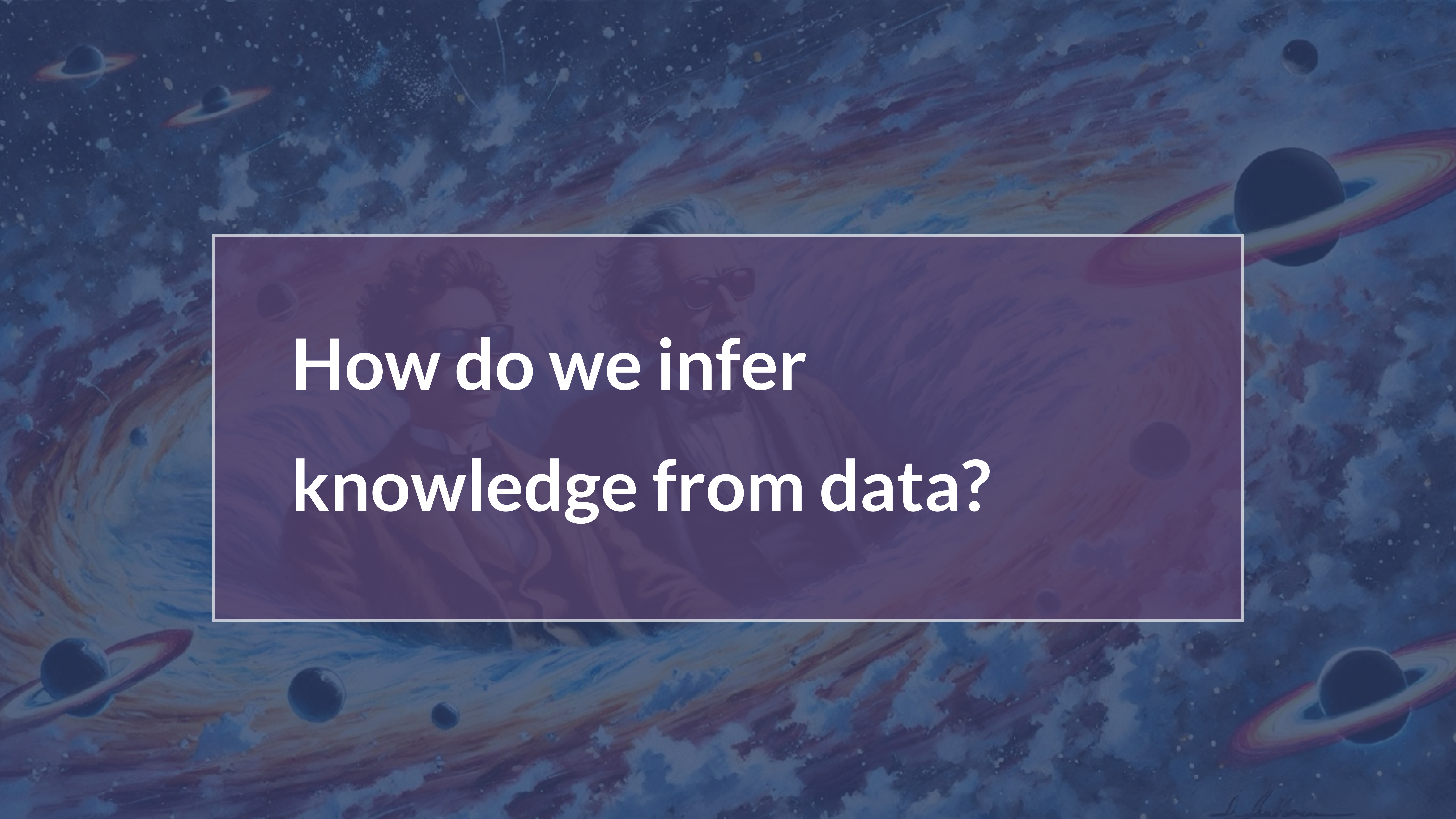
- Clearly declare where and how you used AI on assignments!

What are you hoping to get out of this class?

What will help you get the most out of this class?

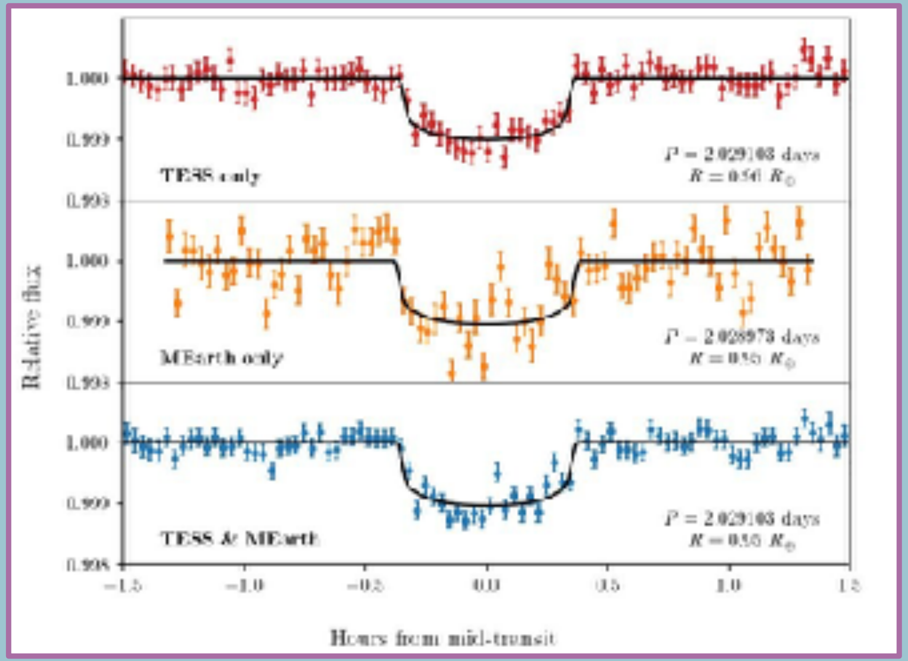
What questions do you have for me about the course?



How do we infer knowledge from data?

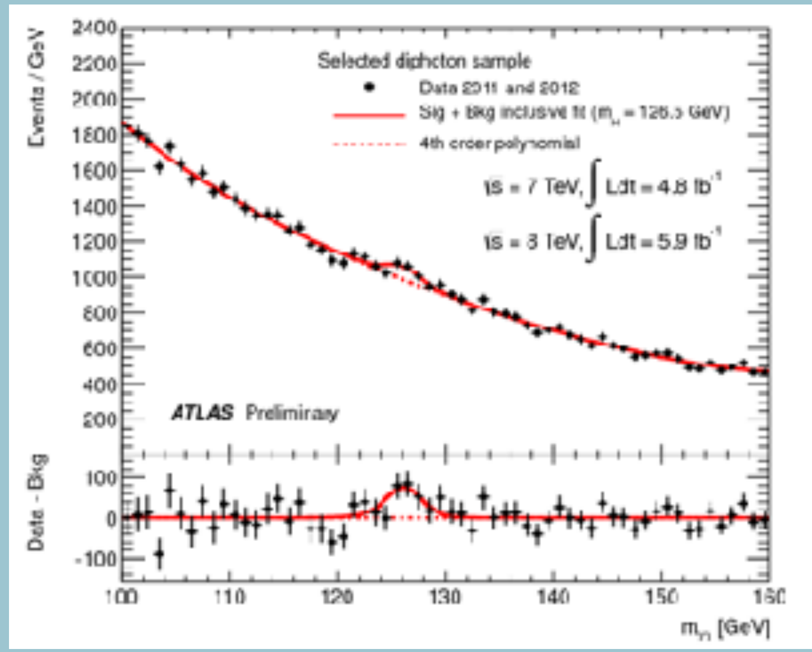
What kinds of questions do we want to ask of our data?

Ment et al (2023)

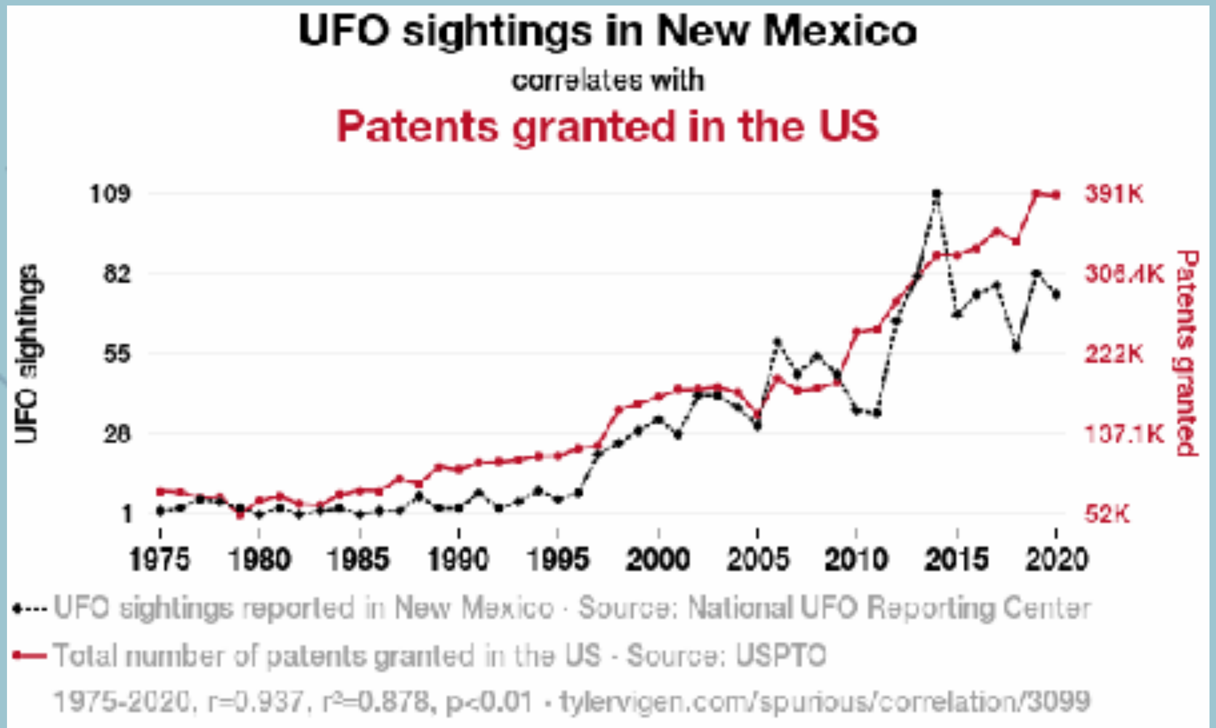


Is there an exoplanet in this system? What are its radius and mass?

Does the Higgs boson exist? What are its properties?



Atlas Collaboration/CERN



https://www.tylervigen.com/spurious/correlation/

Are aliens secretly submitting patent applications?

Sample space

Exploratory Data Analysis

Confirmatory Data Analysis

Do not perform EDA and CDA on the same dataset!

Sample space

Exploratory Data Analysis

Confirmatory Data Analysis

- Visualize and summarise data to extract characteristics

Do **not** perform EDA and CDA on the same dataset!

Sample space

Exploratory Data Analysis

Confirmatory Data Analysis

- Visualize and summarise data to extract characteristics
- Identify patterns

Do not perform EDA and CDA on the same dataset!

Sample space

Exploratory Data Analysis

Confirmatory Data Analysis

- Visualize and summarise data to extract characteristics
- Identify patterns
- Detect anomalies and outliers

Do not perform EDA and CDA on the same dataset!

Sample space

Exploratory Data Analysis

Confirmatory Data Analysis

- Visualize and summarise data to extract characteristics
- Identify patterns
- Detect anomalies and outliers
- Understand relationships between variables

Do not perform EDA and CDA on the same dataset!

Sample space

Exploratory Data Analysis

- Visualize and summarise data to extract characteristics
- Identify patterns
- Detect anomalies and outliers
- Understand relationships between variables
- Open-ended process

Confirmatory Data Analysis

Do not perform EDA and CDA on the same dataset!

Sample space

Exploratory Data Analysis

- Visualize and summarise data to extract characteristics
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Confirmatory Data Analysis

- Test pre-specified hypotheses or models against a dataset

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Sample space

Exploratory Data Analysis

- Visualize and summarise data to extract characteristics
- Identify patterns
- Detect anomalies and outliers
- Understand relationships between variables
- Open-ended process

Confirmatory Data Analysis

- Test pre-specified hypotheses or models against a dataset
- Provide evidence to support or refute the hypothesis or theory

Do not perform EDA and CDA on the same dataset!

Examples

Examples

<u>Exploratory Data Analysis</u>

- Plotting various forms of the data (e.g. histograms, scatter plots)
- Clustering
- Compute summaries of the data (mean, variance, skewness, correlations, …)
- Examples: height, age, time since an event, brightness of a star

<u>Confirmatory Data Analysis</u>

- Hypothesis testing
- Regression, maximum likelihood analysis
- Bayesian inference
- Model comparison

What types of data are there?

How we visualise and summarise data depends on the kind of data we have!

What types of data are there?

How we visualise and summarise data depends on the kind of data we have!

Quantitative

- numerical data: can be discrete or continuous
- Examples: height, age, time since an event, brightness of a star

Categorical

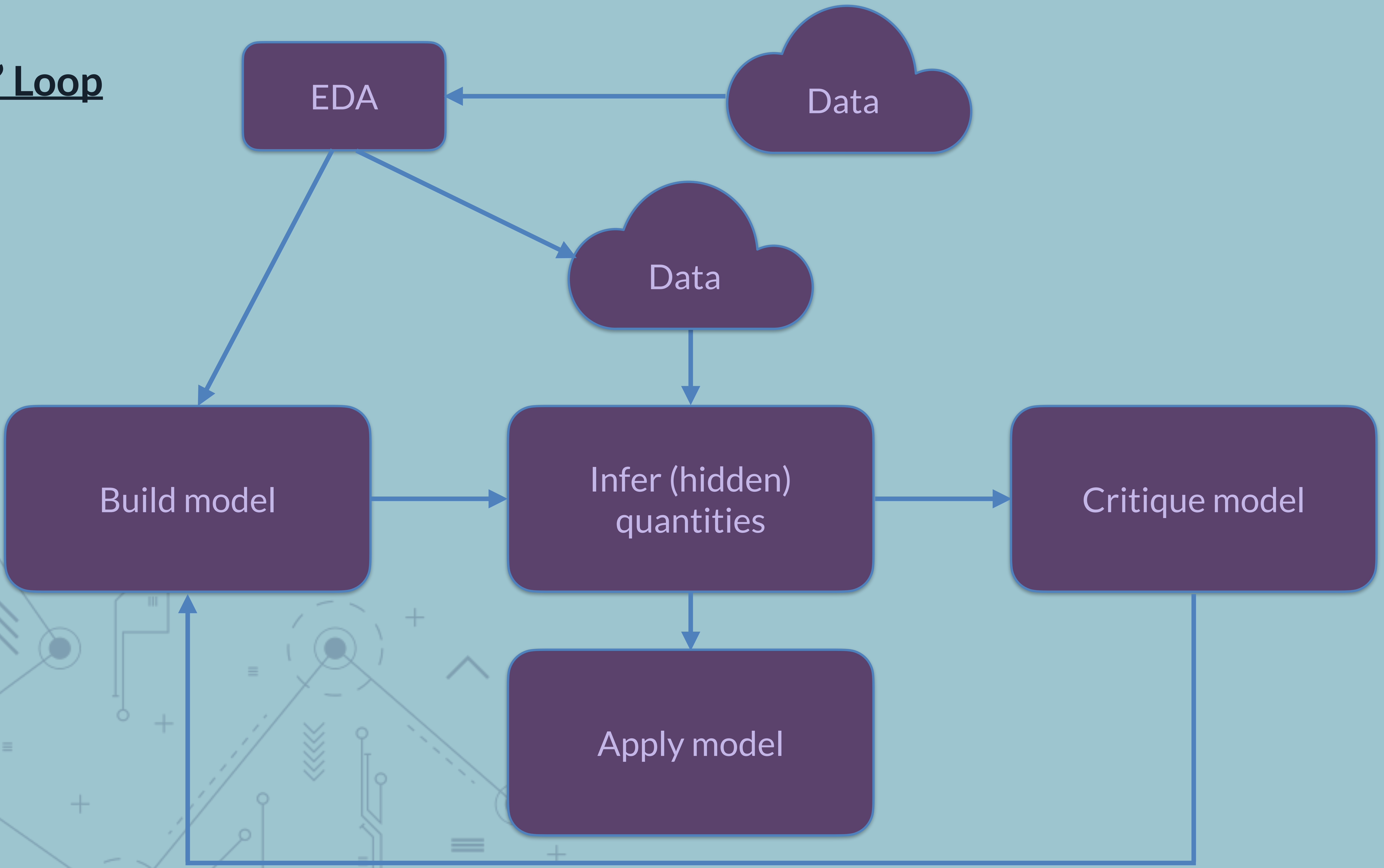
- Can be grouped into a category, type or quality
- Examples: month of birthday, type of galaxy, particle flavour etc.

Ordinal

- have a natural order
- Examples: survey satisfaction ratings, restaurant ratings, grades for coursework

Can you name examples of EDA or CDA?

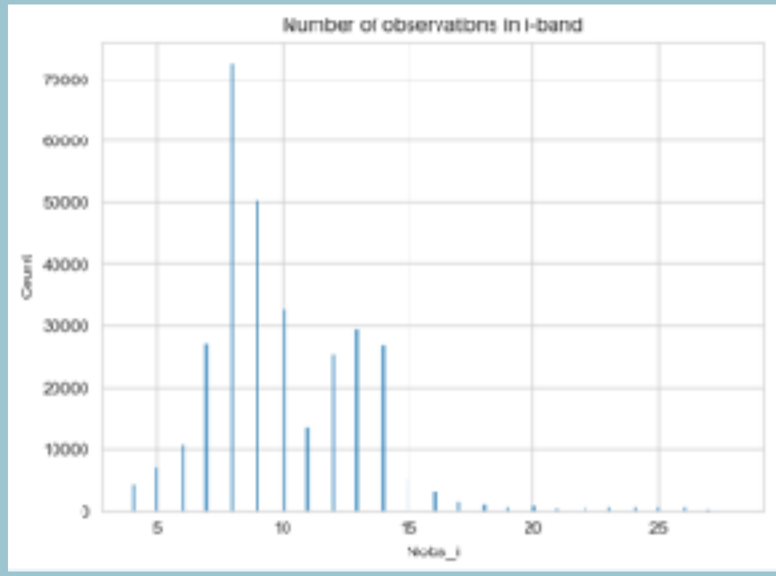
Box' Loop


EDA
Data
Data
Build model
Infer (hidden) quantities
Critique model
Apply model

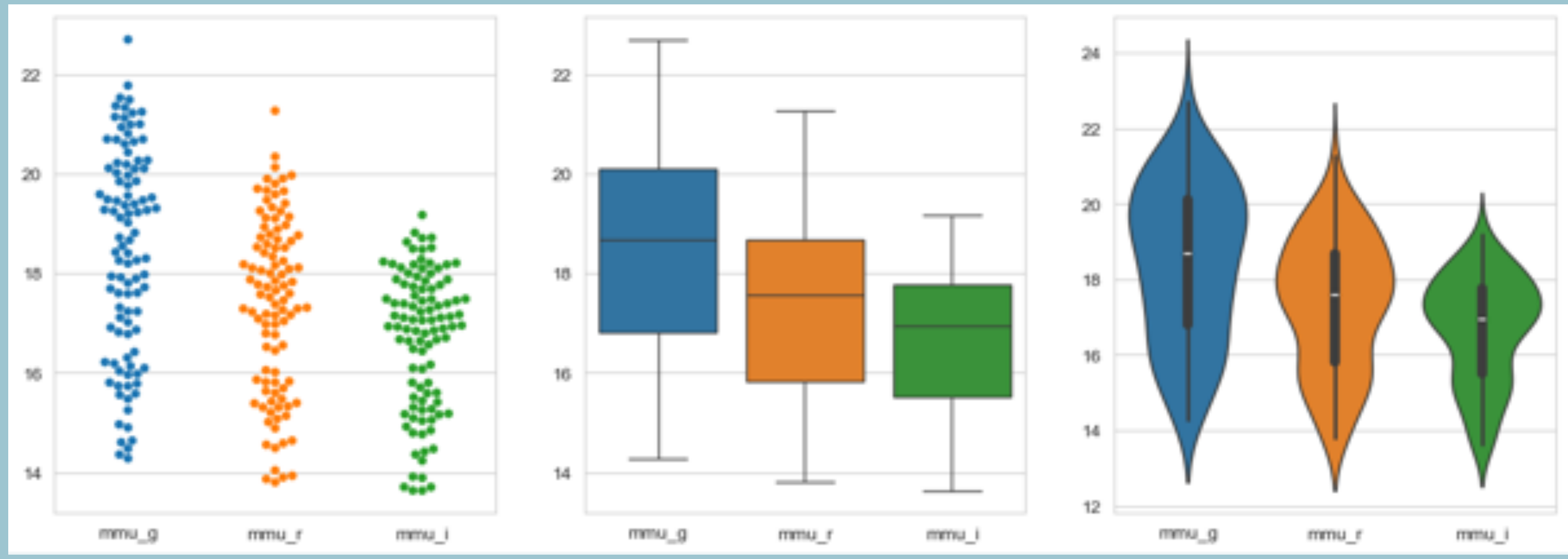
Example based on planet detections —> first planet detection —> Kepler: planets are common —> confirmation/rejection of specific planet formation theories

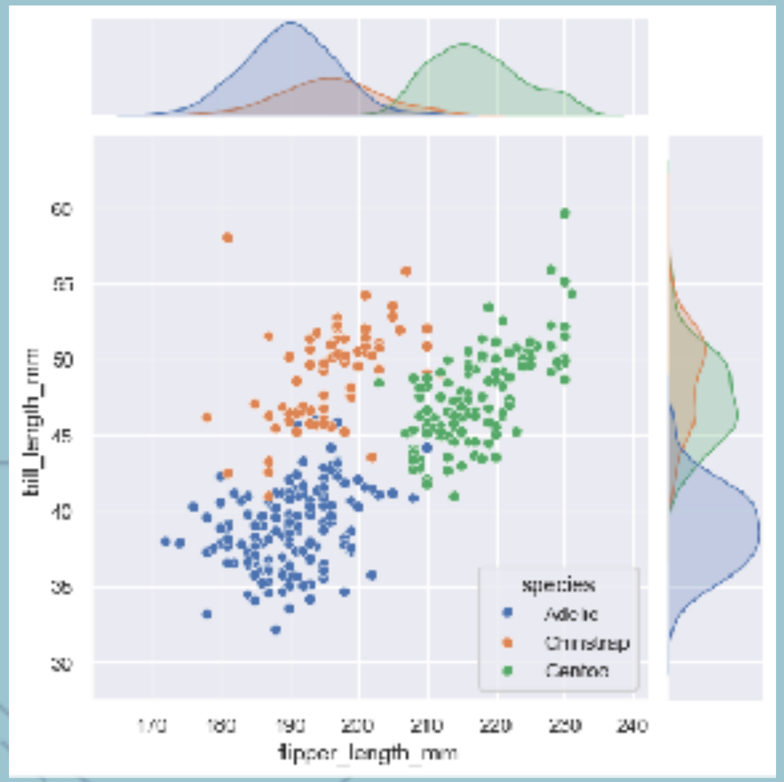
Exploratory Data Analysis: Plotting

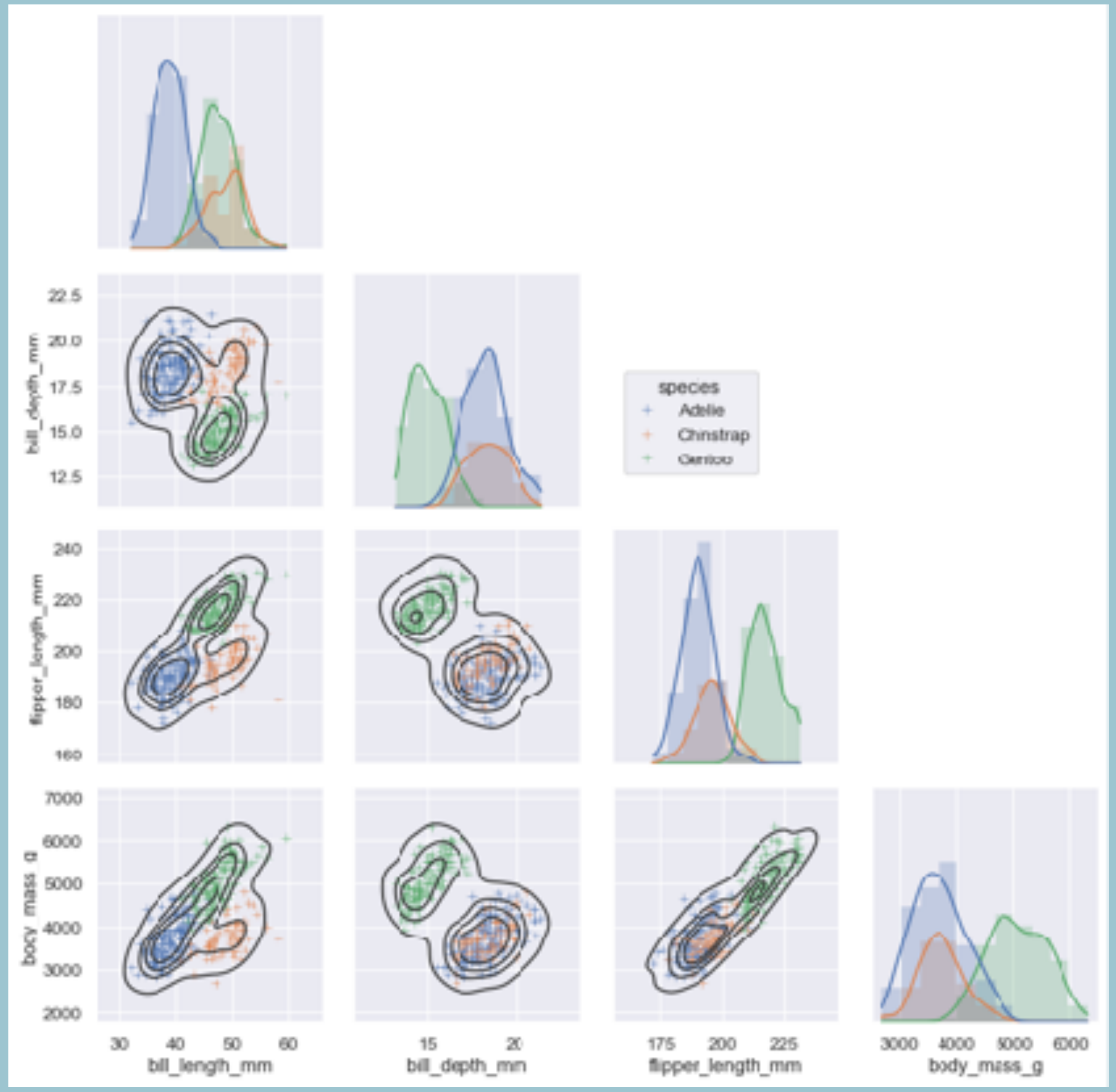
1D discrete



Discrete + continuous







Exploratory Data Analysis: Data summaries

Single variable:

- Geometric mean:
- Median:
- Mode (categorical data):
- variance & standard deviation:
- Interquartile range:
- Minimum/maximum:
- Percentiles:
- Skewness:
- Kurtosis:

Multiple variables

- Covariance:
- Correlation:

Time series summaries:

- Autocorrelation function
- Periodograms/Fourier amplitude spectra:
- Rolling means/variances

Dimensionality reduction

- Principal Component Analysis

…

Exercise:

Planning a Movie Night!

Example: The API Social Committee would like to know whether to schedule a movie night. For this, they’re gathering some information:

[Wooclap: how many movies did you watch last month?]

Brainstorm: what questions can we ask of this data? What other data could we collect that would be helpful?

Based on our data, what can or can’t we conclude about movie watching habits?

Population

Sample

Statistics can be used to understand the underlying population, when all you have is a sample

Beware: representativeness matters a lot!

Population

Sample

- True, underlying distribution for some quantity

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- e.g. the distribution of movies watched per month across the whole world)

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- A sample drawn from some distribution

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Population

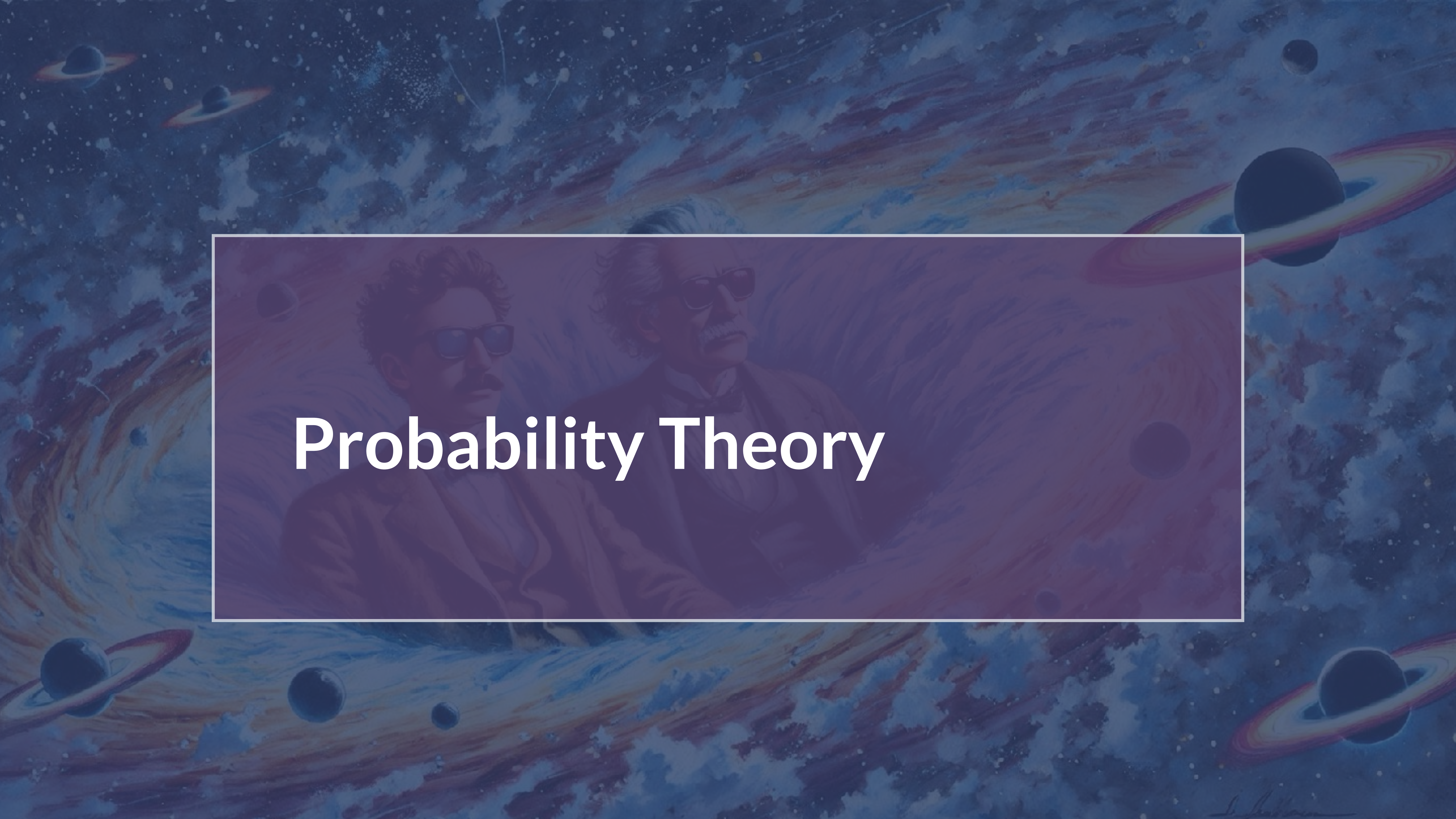
- True, underlying distribution for some quantity
- e.g. the distribution of movies watched per month across the whole world)

Sample

- A sample drawn from some distribution
- e.g. the Physics and Astronomy MSc students in Amsterdam, in January

Statistics can be used to understand the underlying population, when all you have is a sample

Beware: representativeness matters a lot!



Probability Theory

Randomness

Examples:

Randomness

Examples:

Informally: the apparent or actual lack of definite pattern or predictability in information

Mathematically: define randomness and a random variable using the axioms of measure theory

Random variable: a function that assigns a numerical value to each outcome of a random experiment or process

Distribution: tells you the frequency or relative frequency of each possible value/event, or of some data that was collected

Events

Disjoint/mutually exclusive: A and B are disjoint because they don't share any outcomes/ events

Sample space

Sample space

Events

Disjoint/mutually exclusive: A and B are disjoint because they don't share any outcomes/ events

Sample space

Sample space

Example 1: Daniela would like to make it to UvA in time for her lecture.

Event A: the train runs on time

Event B: the train is late

Example 2:

Event A: the train runs on time

Event B: the train is late

Event C+: …

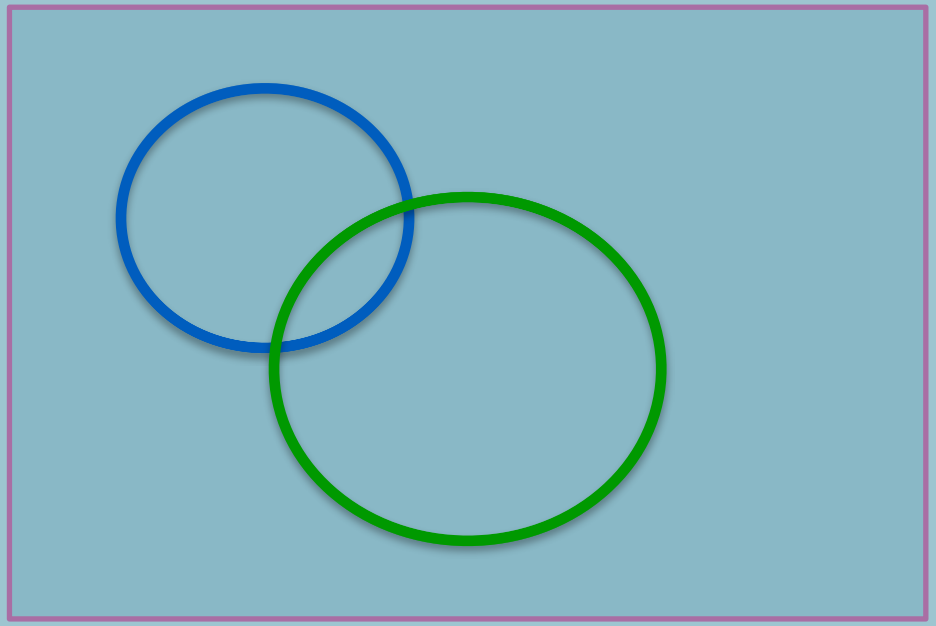
Events

Not disjoint / not mutually exclusive:

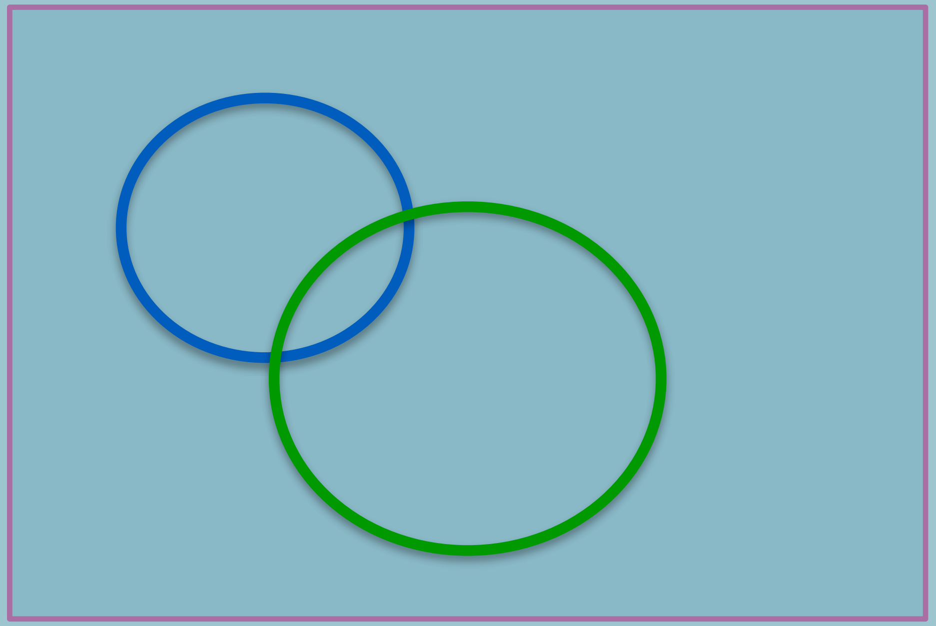
A and B share common outcomes/events

Sample space

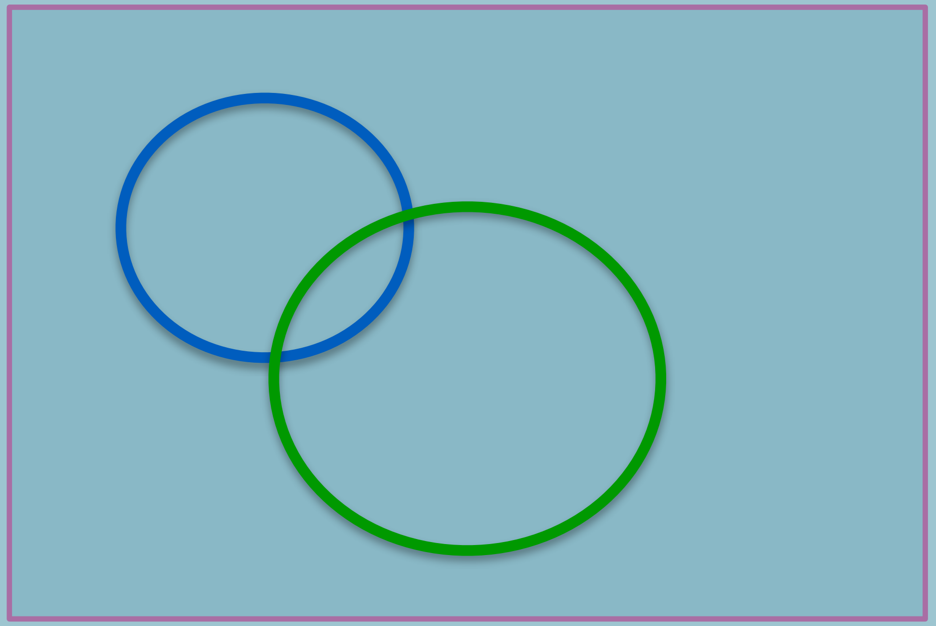
More Examples



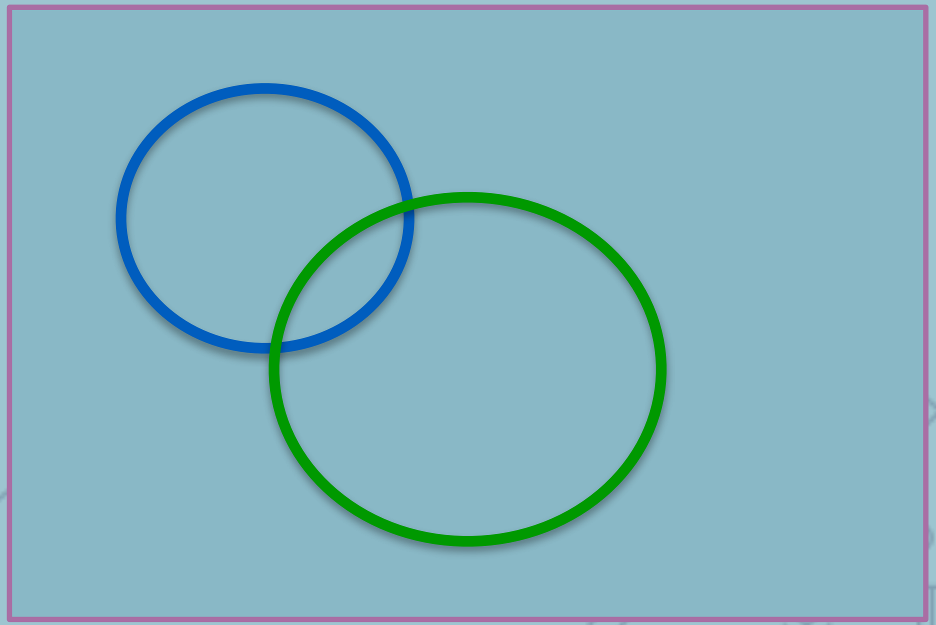
A and B



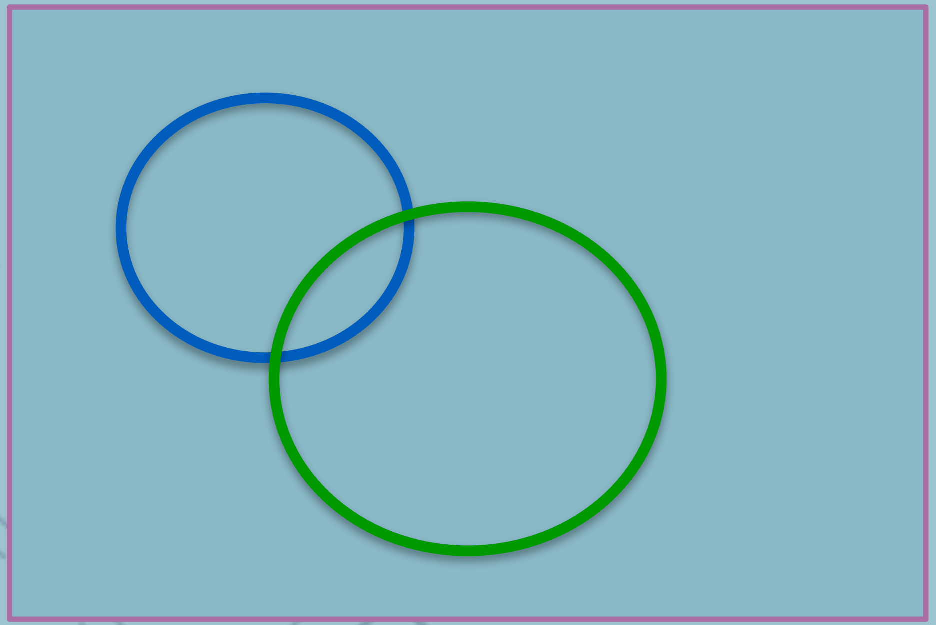
A or B



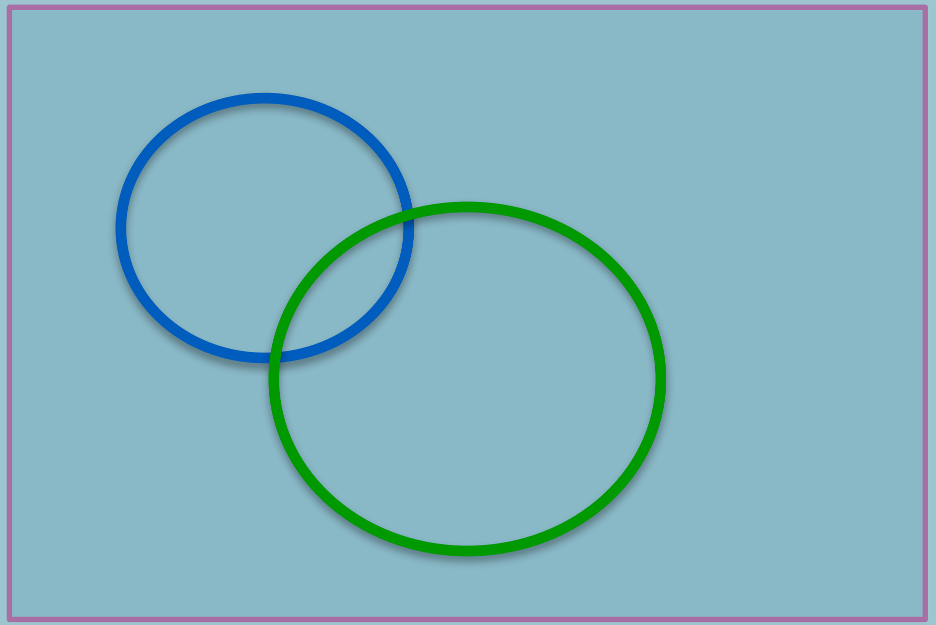
A or B complement



A complement and B



Complement of (A and B)



A complement or B

Check-in

Check-in

A member of the class having watched one, two or three movies last month are

(a) mutually exclusive events

(b) not mutually exclusive events

Probability

Note: there exists more than one formalization of probability.

Most common: Kolmogorov formulation and Cox formulation

Probability is a (real-valued) set function that assigns to each event A in the sample space $\mathbf{S}$ a number $P(A)$, called the probability of the event A, such that the following hold:

1. The probability of any event must be nonnegative, that is, $P(A) \geq 0$.
2. The probability of the sample space is 1, that is, $P(\mathbf{S}) = 1$.
3. Given mutually exclusive events $A_1, A_2, A_3, \ldots$ that is, where $A_i \cap A_j = \emptyset$, for $i \neq j$,
   - the probability of a finite union of the events is the sum of the probabilities of the individual events, that is:
     $P(A_1 \cup A_2 \cup \ldots \cup A_k) = P(A_1) + P(A_2) + \ldots + P(A_k)$

Check-in

Check-in

For our movie exercise …

More Probability

Probability of the Null:

If A is contained in B, then

Probability an event A is between 0 and 1

Probability of the complement of A is

Probability of A or B is

Notation

Probability of S

Probability of A

Probability of A and B

Probability of A or B

Probability of A given B

Probability of not A

Example

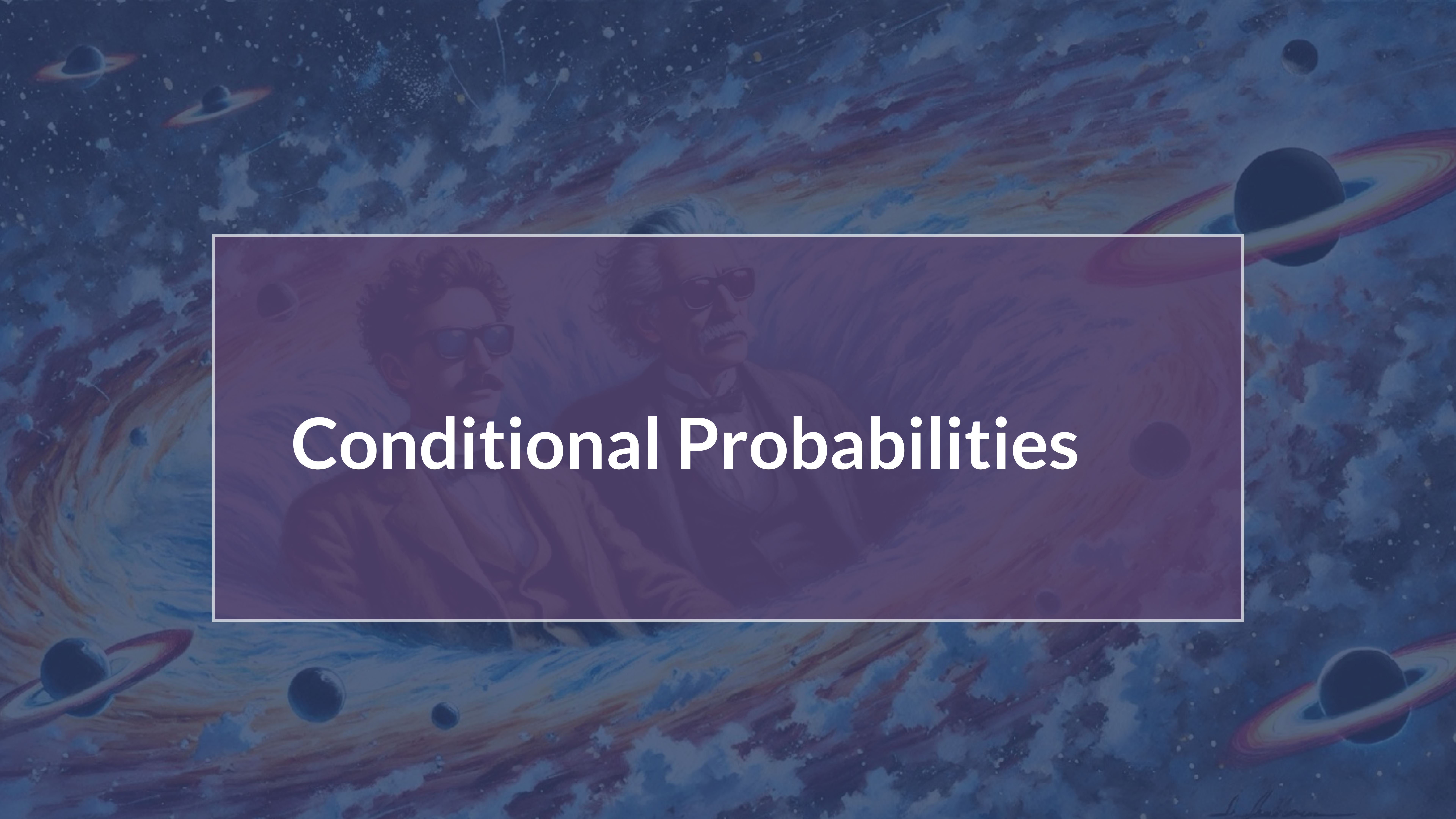
Suppose that this class has the following distribution of students:

Track	Astronomy	GRAPPA	Theoretical Physics	Other	Total
Count	23	11	4	5	43
Proportion	0.53	0.26	0.09	0.12	

1. What is the sample space?

2. What probabilities are associated with each outcome?

3. Are the axioms of probability satisfied?



Conditional Probabilities

Example

Suppose that this class has the following distribution of students:

Track	Astronomy	GRAPPA	Theoretical Physics	Other	Total
Count	23	11	4	5	43
Proportion	0.53	0.26	0.09	0.12	

Conditional Probabilities

- P(Astronomy) = ?
- P(2 | GRAPPA) = ?
- P(TP | 3) = ?
- Is P(1 | Astro) = P(Astro | 1) ?
- $\sum$ P(Other | year) = ?

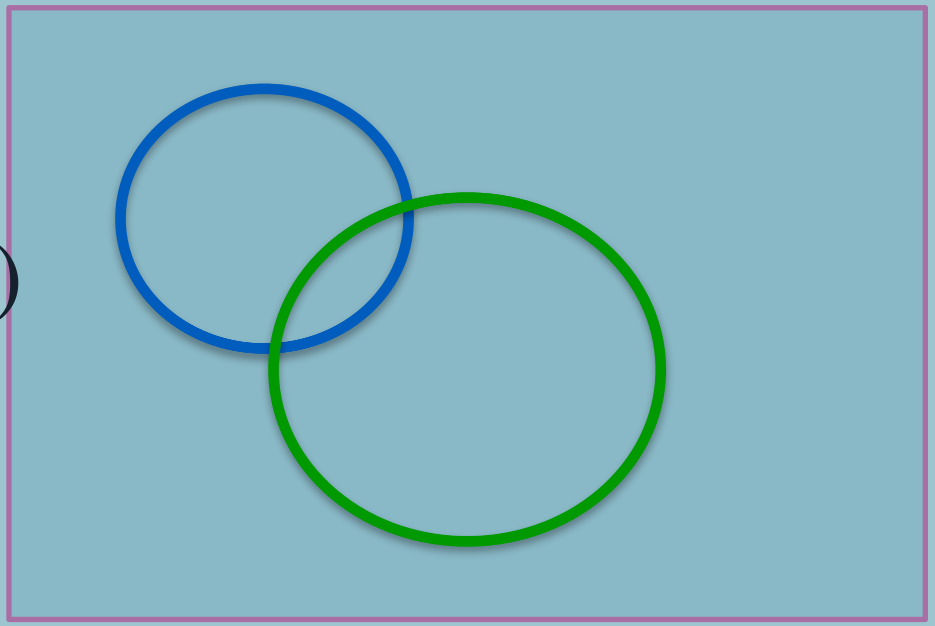
Suppose that this class has the following distribution of students:

Year/Track	Astronomy	GRAPPA	Theoretical Physics	Other	Total
1	19	9	2	3	33
2	3	2	1	0	6
3	1	0	1	2	4
Total	23	11	4	5	43

The probability of an event given some other event:

$$P(B \mid A) = \frac{P(A, B)}{P(A)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Law of total probability

If $A_1, A_2, \ldots, A_k$ are a partition of the sample space, then for any event B

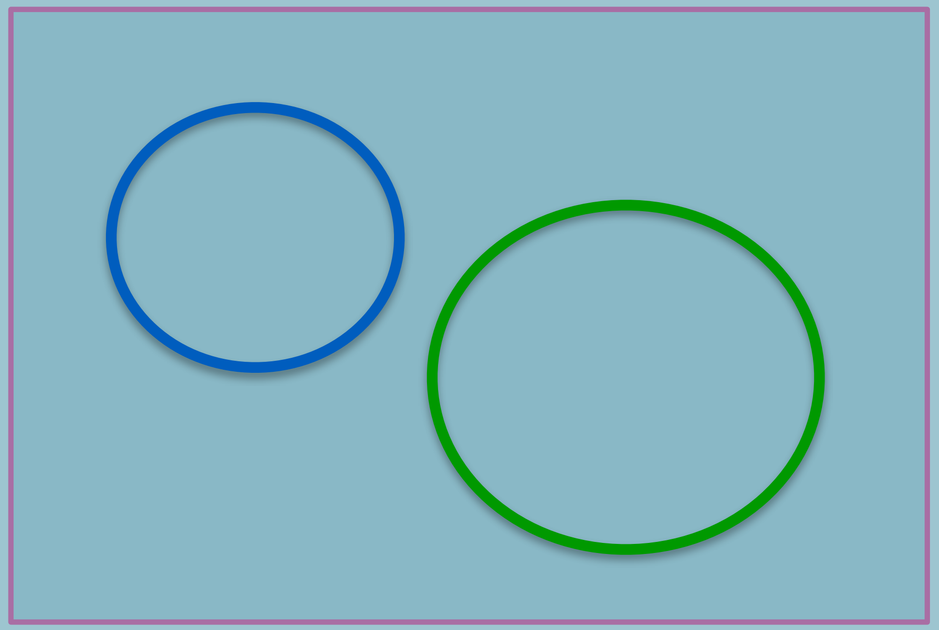
Important: p(☔ | ☁️) $\neq$ p(☁️ | ☔)

Conditional Probabilities

Disjoint/mutually exclusive: probabilities do not share any outcomes

Example: A = astronomy track, B = theoretical physics track

If $P(A) = 0.3$ and $P(B) = 0.25$, what is $P(A \cup B)$?



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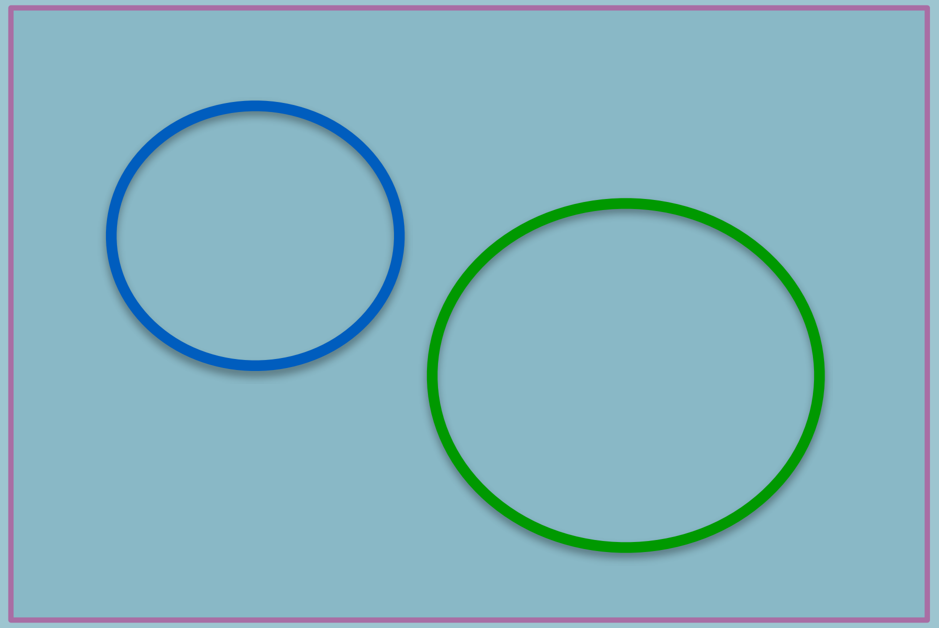

Independence

Events A and B are independent if $P(A, B) = P(A)P(B)$

If A and B are independent, then $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$

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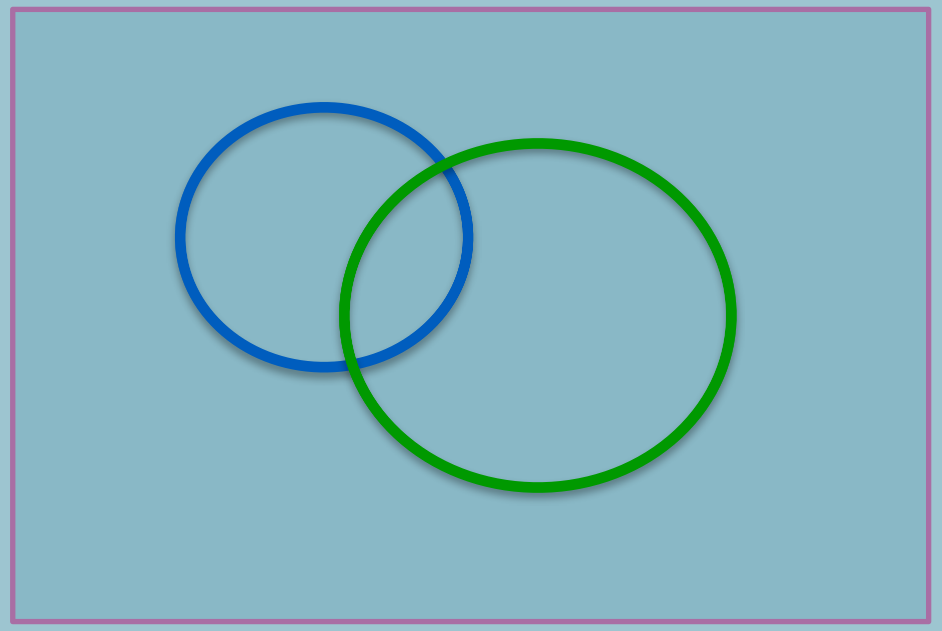

Example: Monty Hall Problem

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

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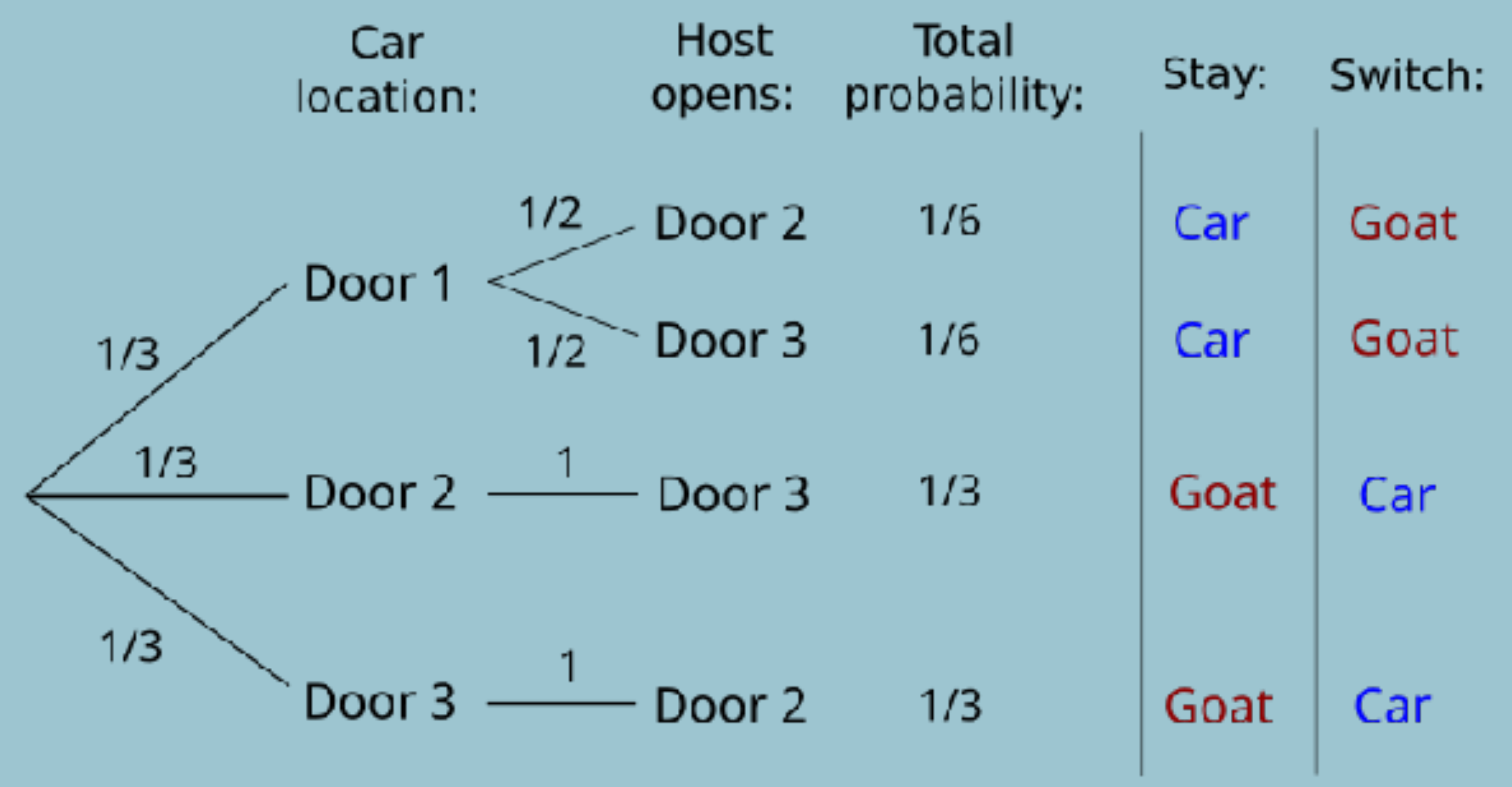

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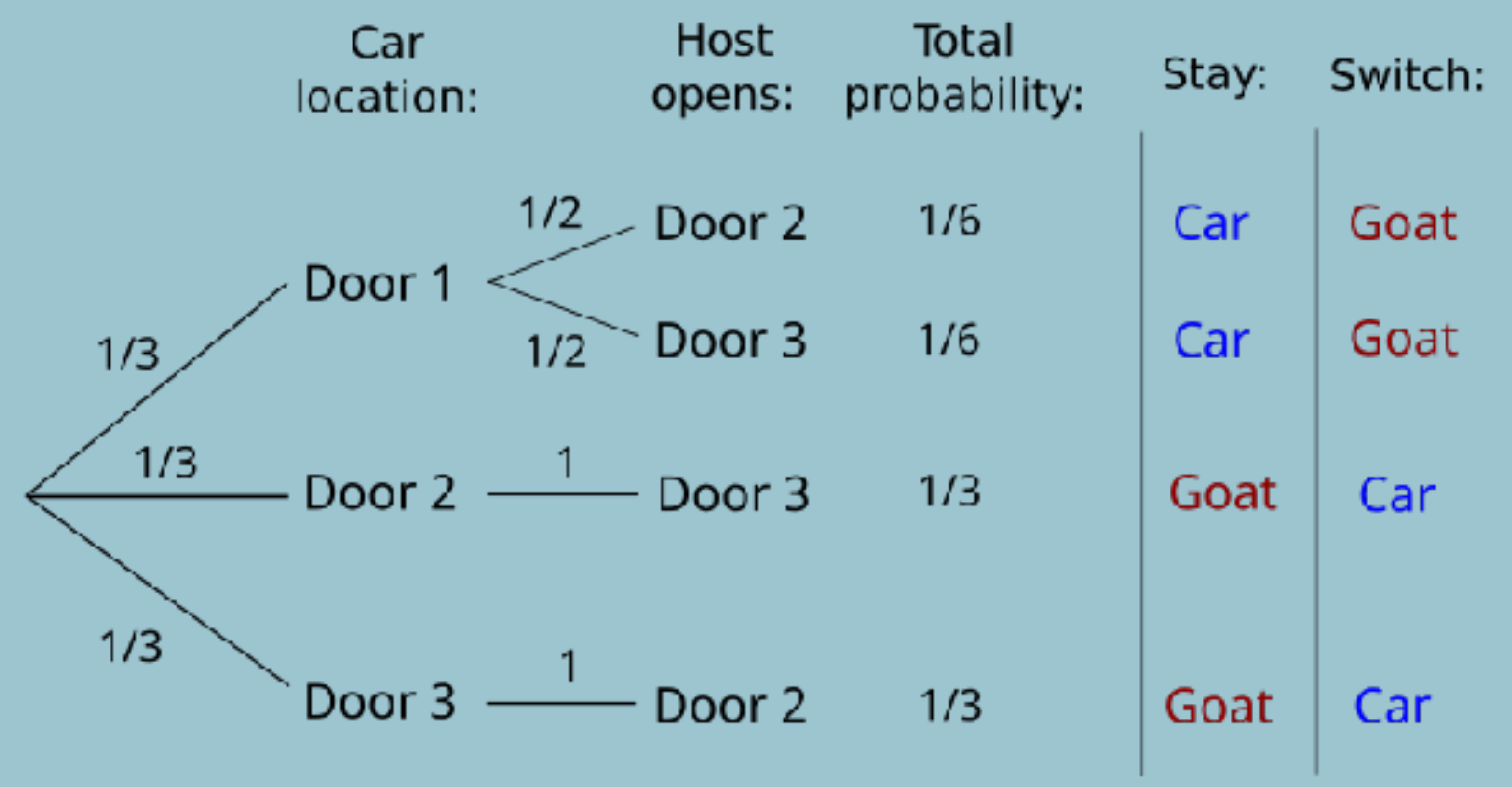

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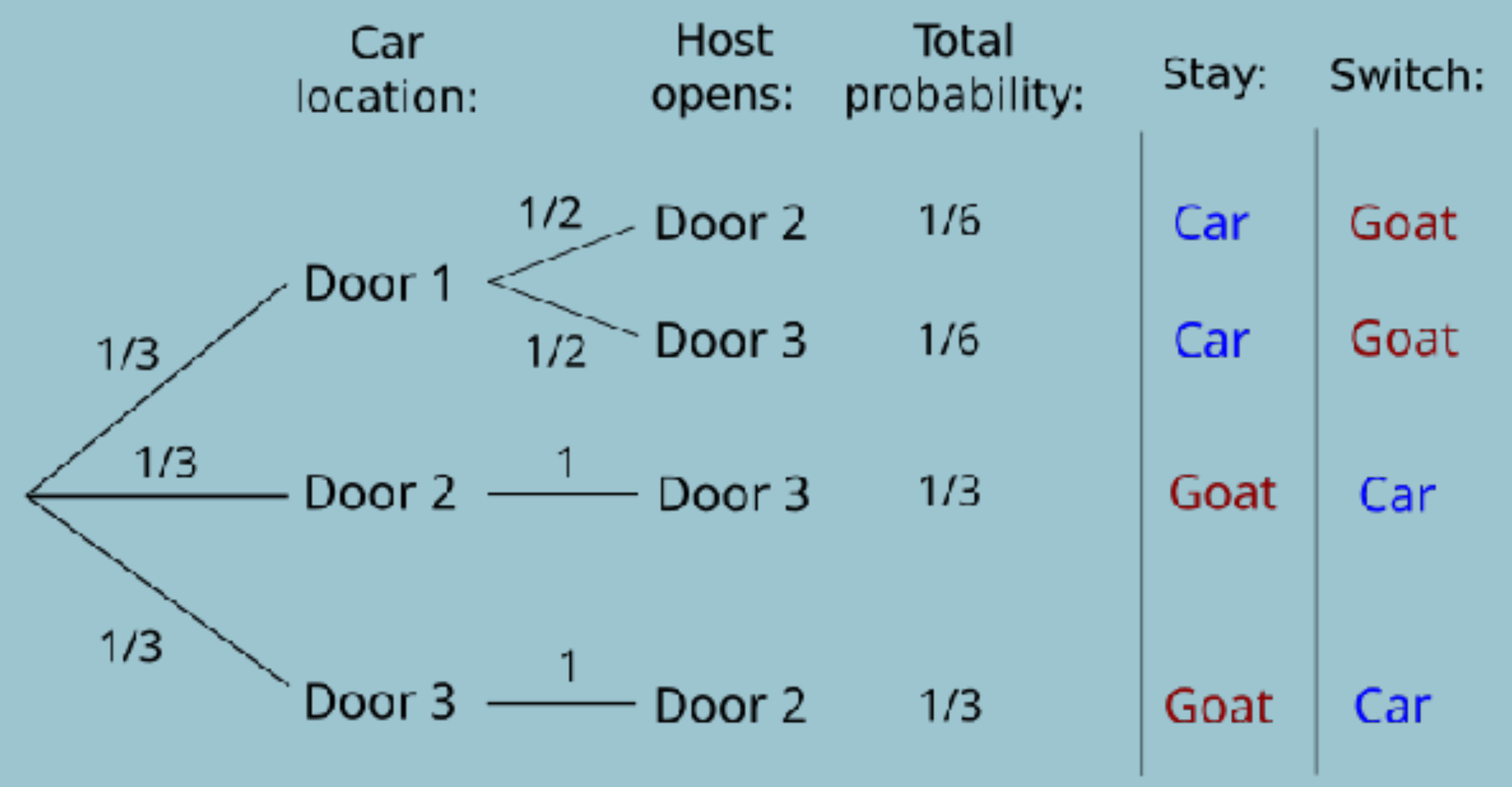

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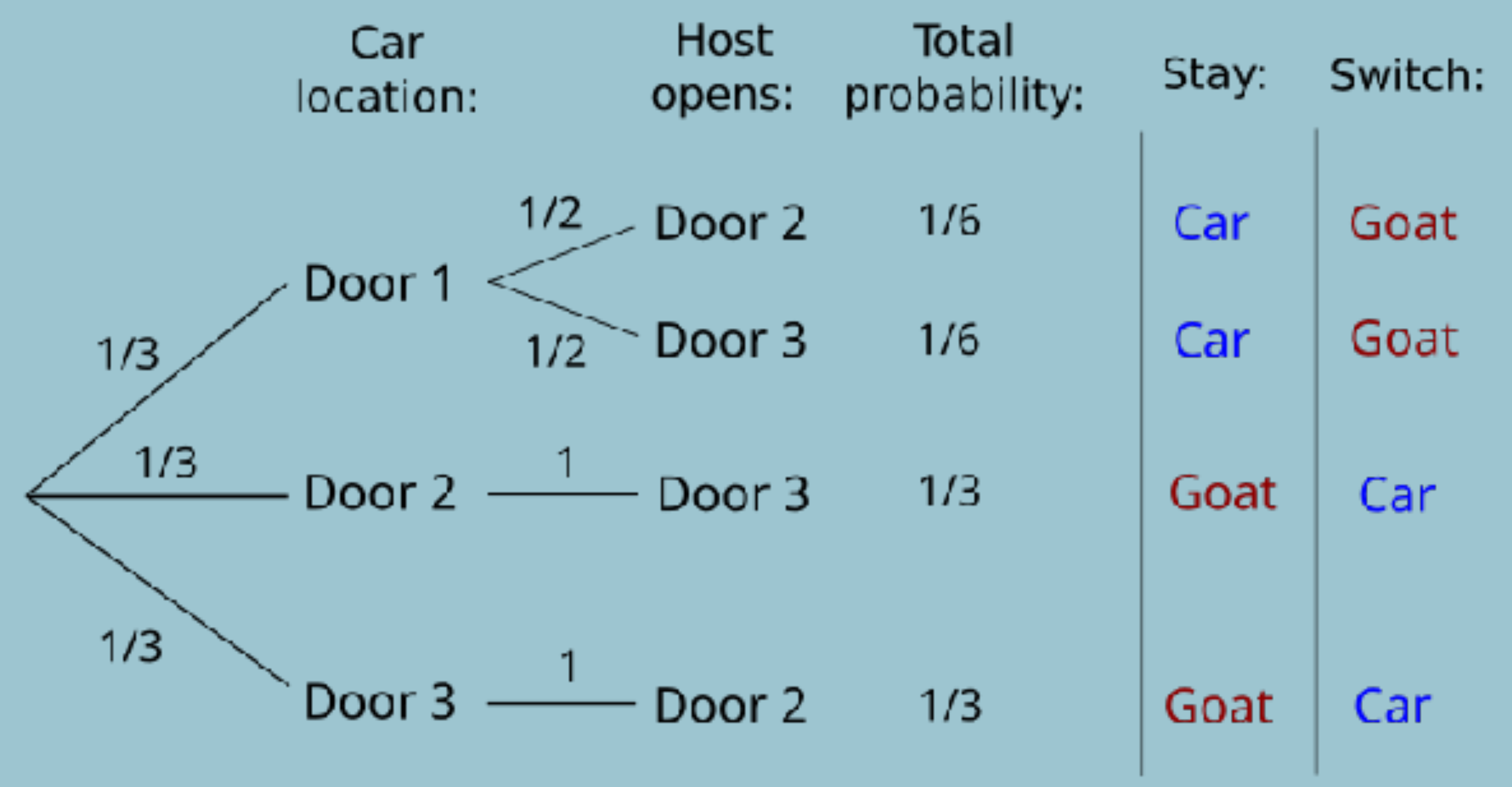

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$$P = 1/3 / (1/3 + 1/6) = 2/3$$

Bayes Theorem

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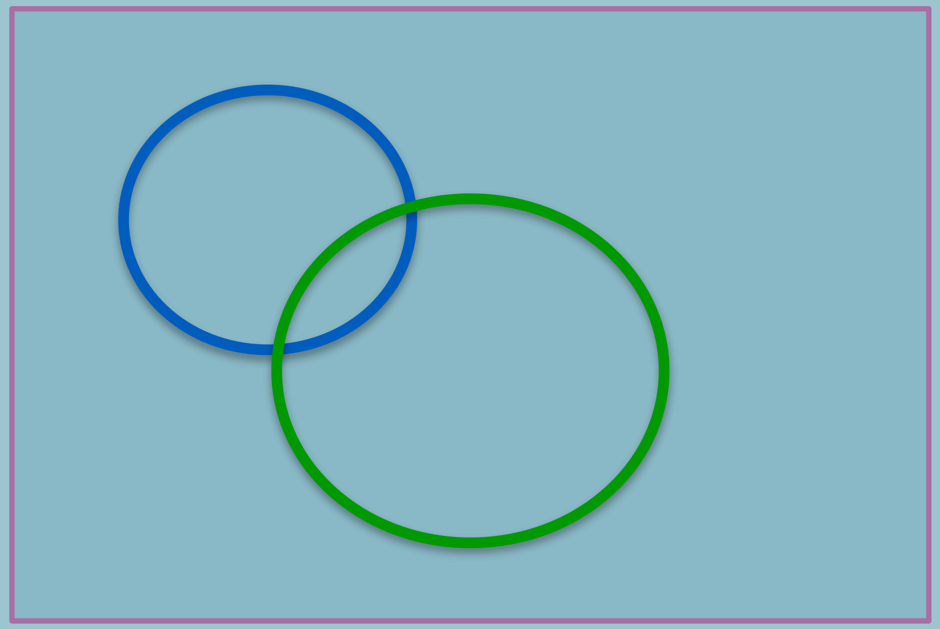

Exercise using Bayes theorem

In a galaxy, 60% of the stellar systems have an Earth-like planet, and all of these systems also have a Jupiter-like planet. However, only half of the systems without an Earth-like planet have a Jupiter-like planet.

What’s the probability that a stellar system with a Jupiter-like planet also has an earth-like planet?

Interpretations of Probability

Discuss with your neighbour:

* How would you define probability?

* Can you think of examples from your daily lives where probability plays a role?

Data

- Imagine we measured the brightnesses of 25 randomly selected stars from the sky

→These data are realizations x of a random variable X that represents the brightness of a star

- To perform (parametric) statistical inference on our data x, we assume how the random variable X is distributed
- Once we have a statistical model for how X is distributed, then we can say things like “the probability of observing a star with magnitude less than 17 is …”

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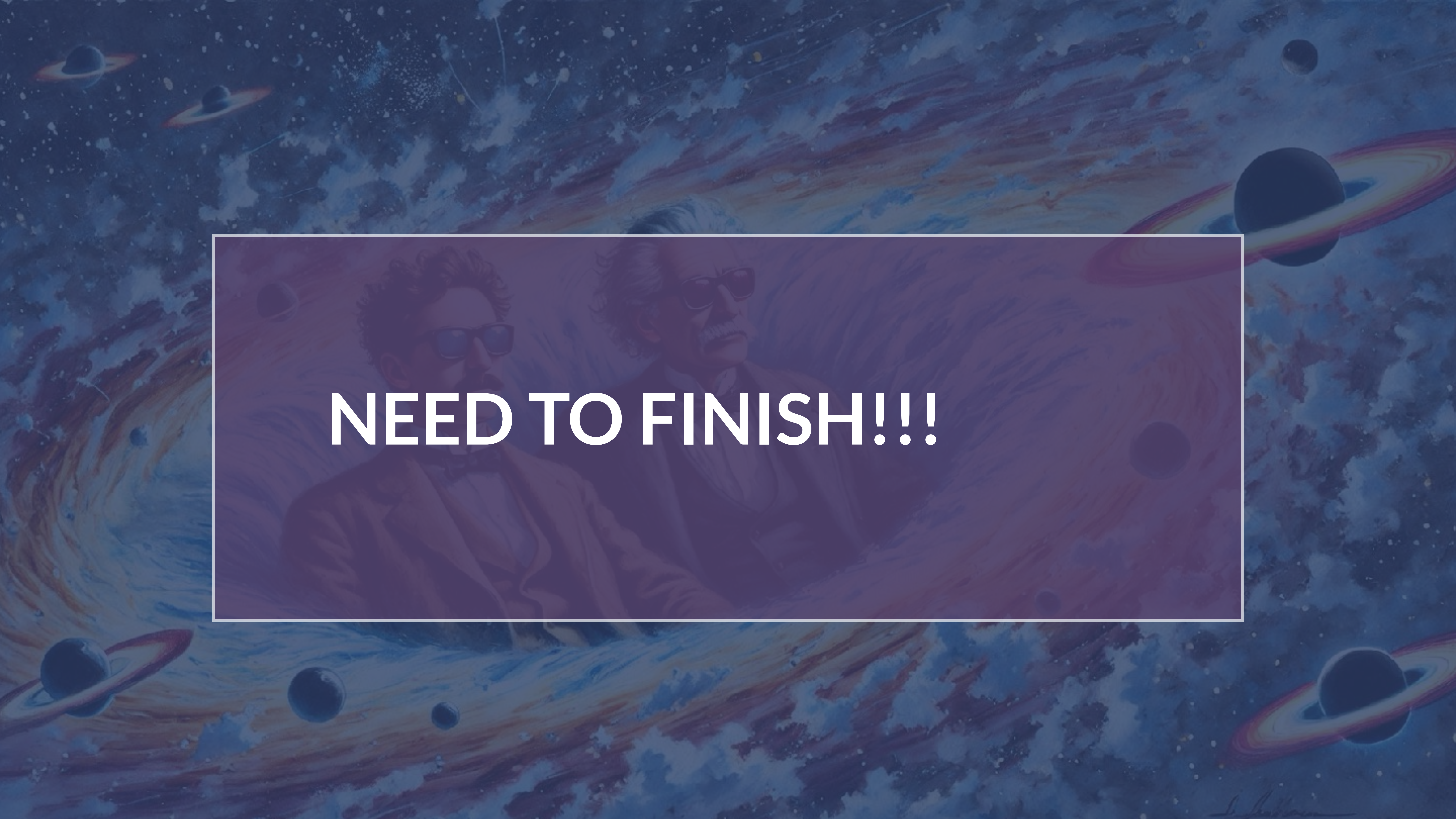
- Our belief in an outcome is a combination of our prior beliefs in that outcome, and the information supplied by the data.
- Enables direct computation of probabilities of hypotheses



NEED TO FINISH!!!

Tutorial

Example Exam Questions I

A Gamma-Ray Burst (GRB) catalogue contains 5,000 events. For each GRB, we have recorded the total energy produced, its duration and its light curve. Imagine that we have a physical model that predicts GRB duration based on its progenitor type and energy (compact object mergers versus supernova).

Give one example of an EDA step and one example of a CDA question.

Solution:
EDA: Plot histograms of durations (T90), scatter plots of duration vs. fluence, look for clusters.

CDA: compare predictions from the two models to the observed data, e.g. via regression.

Example Exam Questions II

Classify each variable into its data type.
a) Photon counts per second.
b) Stellar spectral type (O, B, A…).
c) FRB confidence ranking (“unlikely”, “possible”, “likely”).
d) Galaxy redshift.

Solution:

a) Quantitative (discrete).
b) Categorical (nominal).
c) Ordinal (ranked categories).
d) Quantitative (continuous).

Example Exam Questions III

From archival data:

- 40% of blazars exhibit strong X-ray variability.
- 70% of gamma-ray loud blazars show strong X-ray variability.
- 30% of all blazars are gamma-ray loud.
  Compute the probability that a randomly selected blazar is **gamma-ray loud and shows strong X-ray variability**.

Assignment Goals

* Practice statistical concepts with real-world data
* Get experience with real-world data analysis workflows
* Excellent preparation for your MSc thesis
* Also very useful for jobs outside of academia!

AI Use in the Assignments

add

Example: Debugging

Add a slide on effective debugging

Where to get help

Check the course lecture notes!!!
They have example code!!